



# Finanzas en R

## Actividad 03

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### Actividad 3

#### R y SQL

Veremos un ejemplo simple de conexión entre R y SQL; en este caso utilizamos SQLite.

#### Librerías y conexión

```
library(RSQLite)
library(dplyr)
```

Se establece la conexión con:

```
con <- dbConnect(SQLite())
```

#### Escritura de tabla y schema

Escribimos una tabla en base al set de prueba `mtcars`

```
dbWriteTable(con, "mtcars_sql", mtcars)
```

Dos maneras para inspeccionar el schema y las tablas son los siguientes:

```
tables <- dbListTables(con)

schema <- dbGetQuery(con, "PRAGMA table_info(mtcars_sql)")
```

El primero nos permite ver las tablas en el schema en donde estamos trabajando y el segundo nos permite inspeccionar las características de la tabla especificada. ## Lectura usando SQL

Una primera manera es utilizar la query relacionada con la conexión de SQL que estamos utilizando:

```
data_01 <- dbGetQuery(con, 'SELECT MPG, CYL FROM mtcars_sql WHERE MPG > 30')

data_01
```

|   | mpg  | cyl |
|---|------|-----|
| 1 | 32.4 | 4   |
| 2 | 30.4 | 4   |
| 3 | 33.9 | 4   |
| 4 | 30.4 | 4   |

Otra forma es “estorear” la query para después hacer la búsqueda:

```
get_data_02 <- dbSendQuery(con, "SELECT MPG, CYL FROM mtcars_sql WHERE MPG > 30")

data_02 <- dbFetch(get_data_02)

data_02
```

|   | mpg  | cyl |
|---|------|-----|
| 1 | 32.4 | 4   |
| 2 | 30.4 | 4   |
| 3 | 33.9 | 4   |
| 4 | 30.4 | 4   |

La otra manera es convertir la tabla de sql a un dataframe en base a dplyr. Después usamos los comandos propios de R para lograr el mismo resultado anterior:

```
tabla_cars <- tbl(con, 'mtcars_sql')
```

Warning: Closing open result set, pending rows

```
tabla_cars %>%  
  select(mpg, cyl) %>%  
  filter(mpg>30) %>%  
  arrange(mpg)
```

```
# Source:      SQL [?? x 2]  
# Database:    sqlite 3.50.1 []  
# Ordered by: mpg  
   mpg    cyl  
<dbl> <dbl>  
1  30.4     4  
2  30.4     4  
3  32.4     4  
4  33.9     4
```

¿Cómo se vería esto en una query? Preguntemosle a R:

```
tabla_cars %>%  
  select(mpg, cyl) %>%  
  filter(mpg>30) %>%  
  arrange(mpg) %>%  
  show_query()
```

```
<SQL>  
SELECT `mpg`, `cyl`  
FROM `mtcars_sql`  
WHERE (`mpg` > 30.0)  
ORDER BY `mpg`
```

## Obtener datos financieros

### Ejemplo 1

```
# Load necessary libraries  
library(quantmod)  
library(DBI)  
library(RSQLite)
```

```

# Define a function to fetch financial data
fetch_financial_data <- function(symbol, start_date, end_date) {
  # Fetch data using quantmod
  data <- getSymbols(symbol, from = start_date, to = end_date, auto.assign = FALSE)
  return(data)
}

# Define your SQLite database file
sqlite_file <- "financial_data.db"

# Connect to SQLite database
conn <- dbConnect(SQLite(), sqlite_file)

# Define the symbol and time frame
symbol <- "AAPL"
start_date <- "2020-01-01"
end_date <- Sys.Date() # Today's date

# Fetch financial data
financial_data <- fetch_financial_data(symbol, start_date, end_date)

# Convert xts object to data frame
financial_data_df <- data.frame(date = index(financial_data), coredata(financial_data))

# Write data to SQLite database
dbWriteTable(conn, "stock_data_01", financial_data_df, overwrite = TRUE)

# Close the database connection

data_financial <- dbGetQuery(conn, 'SELECT * FROM stock_data')

data_financial %>% head()

```

|   | date  | AAPL.Open | AAPL.High | AAPL.Low | AAPL.Close | AAPL.Volume | AAPL.Adjusted |
|---|-------|-----------|-----------|----------|------------|-------------|---------------|
| 1 | 18263 | 74.0600   | 75.1500   | 73.7975  | 75.0875    | 135480400   | 73.05943      |
| 2 | 18264 | 74.2875   | 75.1450   | 74.1250  | 74.3575    | 146322800   | 72.34914      |
| 3 | 18267 | 73.4475   | 74.9900   | 73.1875  | 74.9500    | 118387200   | 72.92564      |
| 4 | 18268 | 74.9600   | 75.2250   | 74.3700  | 74.5975    | 108872000   | 72.58267      |
| 5 | 18269 | 74.2900   | 76.1100   | 74.2900  | 75.7975    | 132079200   | 73.75025      |
| 6 | 18270 | 76.8100   | 77.6075   | 76.5500  | 77.4075    | 170108400   | 75.31676      |

Reflexionemos sobre el paso a paso.

## Ejemplo 2

```
# Load necessary libraries
library(quantmod)
library(DBI)
library(RSQLite)

# Define a function to fetch financial data
fetch_financial_data <- function(symbol, start_date, end_date) {
  # Fetch data using quantmod
  data <- getSymbols(symbol, from = start_date, to = end_date, auto.assign = FALSE)
  return(data)
}

# Define your SQLite database file
sqlite_file <- "financial_data.db"

# Connect to SQLite database
conn <- dbConnect(SQLite(), sqlite_file)

# Define the symbol and time frame
symbol <- "AAPL"
start_date <- "2020-01-01"
end_date <- Sys.Date() # Today's date

# Fetch financial data
financial_data <- fetch_financial_data(symbol, start_date, end_date)

# Convert xts object to data frame
financial_data_df <- data.frame(date = index(financial_data), coredata(financial_data))

# Define SQL command to create table
create_table_sql <- "
CREATE TABLE IF NOT EXISTS stock_data (
  date TEXT PRIMARY KEY,
  open REAL,
  high REAL,
  low REAL,
  close REAL,
```

```

        volume REAL,
        adjusted REAL
    );
"

# Execute SQL command to create table
dbExecute(conn, create_table_sql)

```

[1] 0

```

# Prepare data for insertion
insert_values_sql <- "INSERT OR REPLACE INTO stock_data VALUES (?, ?, ?, ?, ?, ?, ?)"

# Insert data into table row by row
for (i in 1:nrow(financial_data_df)) {
  row_values <- unname(as.list(financial_data_df[i, ]))
  dbExecute(conn, insert_values_sql, params = row_values)
}

dbWriteTable(conn, "stock_data_02", financial_data_df, overwrite = TRUE)

financial_data_df %>% head()

```

|   | date       | AAPL.Open | AAPL.High | AAPL.Low | AAPL.Close | AAPL.Volume | AAPL.Adjusted |
|---|------------|-----------|-----------|----------|------------|-------------|---------------|
| 1 | 2020-01-02 | 74.0600   | 75.1500   | 73.7975  | 75.0875    | 135480400   | 72.27156      |
| 2 | 2020-01-03 | 74.2875   | 75.1450   | 74.1250  | 74.3575    | 146322800   | 71.56894      |
| 3 | 2020-01-06 | 73.4475   | 74.9900   | 73.1875  | 74.9500    | 118387200   | 72.13921      |
| 4 | 2020-01-07 | 74.9600   | 75.2250   | 74.3700  | 74.5975    | 108872000   | 71.79993      |
| 5 | 2020-01-08 | 74.2900   | 76.1100   | 74.2900  | 75.7975    | 132079200   | 72.95492      |
| 6 | 2020-01-09 | 76.8100   | 77.6075   | 76.5500  | 77.4075    | 170108400   | 74.50454      |

Veamos de nuevo el schema:

```

tables <- dbListTables(conn)
tables

```

[1] "stock\_data" "stock\_data\_01" "stock\_data\_02"

Hacemos un gráfico simple:

```

# Load necessary libraries
library(ggplot2)

# Convert xts object to data frame
financial_data_df <- data.frame(date = index(financial_data), coredata(financial_data))

# Convert date to Date class
financial_data_df$date <- as.Date(financial_data_df$date)

# Plot line chart
ggplot(financial_data_df, aes(x = date)) +
  geom_line(aes(y = AAPL.Close, color = "Close")) +
  geom_line(aes(y = AAPL.Open, color = "Open")) +
  geom_line(aes(y = AAPL.High, color = "High")) +
  geom_line(aes(y = AAPL.Low, color = "Low")) +
  scale_color_manual(values = c("Close" = "blue", "Open" = "red", "High" = "green", "Low" = "orange")) +
  labs(x = "Date", y = "Price", title = "Apple Share Prices") +
  theme_minimal()

```

